Predicting Port Congestion and Shipment Delays

Technical report · Surya Pendekatla · Aiqon Jr AI Engineer take-home · November 2026 · Code: github.com//port-congestion-predictor

Summary. The system predicts, **48 hours before a ship arrives**, whether it will wait **more than 24 hours** at anchorage for a berth, and returns a probability, a confidence level, the factors behind it in plain English, and a recommended action. On a held-out test period covering monsoon and peak season (May–Oct 2026, 526 port calls), it catches **81% of delays at 79% precision** (ROC-AUC 0.92) and cuts the cost of wrong calls by **17%** versus a planner's rule of thumb. The data is **synthetic**, produced by an hour-by-hour port simulation; Section 7 sets out what that does and does not prove.

1. Problem definition: what counts as congestion

The unit of prediction is one **port call** (one ship visiting one port). A call is **delayed** if the ship waits **more than 24 h at anchorage** before getting a berth. Waiting time is used instead of total turnaround because turnaround also includes unloading time, which depends on cargo volume, not congestion. 24 h is the point where delivery dates, truck bookings and onward connections start to break. The prediction is made **48 h before arrival**: early enough to act (slow down, warn the customer, re-plan trucks), late enough that the port situation is reasonably known. Both numbers are single settings in *src/config.py*.

2. Data: how delays were simulated and how ground truth was defined

If the label were written as a formula of the features (e.g. *delay = 3×queue + 2×storm*), any model would just rediscover that formula and prove nothing. Instead, *src/simulate.py* runs **an hour-by-hour queue simulation** of four fictional ports. Ships arrive, wait at anchorage (first come, first served) until a suitable berth is free and the port is open, are unloaded, and leave. **Waiting time comes out of this process; it is never typed in.** The drivers built into the simulation are:

- **Arrivals:** random (Poisson) with seasonality (Aug–Oct peak, Feb lull), a weekday pattern, and slowly drifting **"waves"** that bunch ships together, as happens after upstream disruptions.
- **Capacity:** 2–4 berths per port; ships over 8,000 TEU need one of the few deep-water berths; planned maintenance removes a berth for 3–10 days; unloading time grows with ship size.
- **Weather:** hourly wind, more storms in monsoon (Jun–Sep) and cyclone season (Oct–Nov). The port stops above 45 km/h and cranes run at half speed between 35 and 45 km/h.
- **Labour:** strikes of 12–72 h; 70% announced in advance, **30% sudden** (unpredictable on purpose).
- **Port differences:** a busy hub (Northbay), a storm-exposed port (Eastgate), a small strike-prone port with **daylight-only berthing** (Southreach) and a quiet port (Westhaven).
- **Messy data, injected deliberately:** weather forecast missing 8%, incoming-traffic (AIS) data missing 6%, vessel size missing 3%; ship ETAs off by ~8 h; 2-day wind forecast off by ~7 km/h.

Ground truth: for each simulated call we store (a) only what was knowable 48 h before arrival and (b) the simulated waiting time, with *delayed = wait > 24 h*. Outcome columns are never used as model inputs. Result: **2,064 port calls over 24 months** (Nov 2024–Oct 2026), 29% delayed (by port: Northbay 22%, Eastgate 38%, Southreach 31%, Westhaven 22%). The brief suggests 100–500 rows. More were generated because a queue needs realistic traffic to produce congestion, and the learning curve shows 500 rows already reaches ROC-AUC 0.916 versus 0.924 with all 1,537.

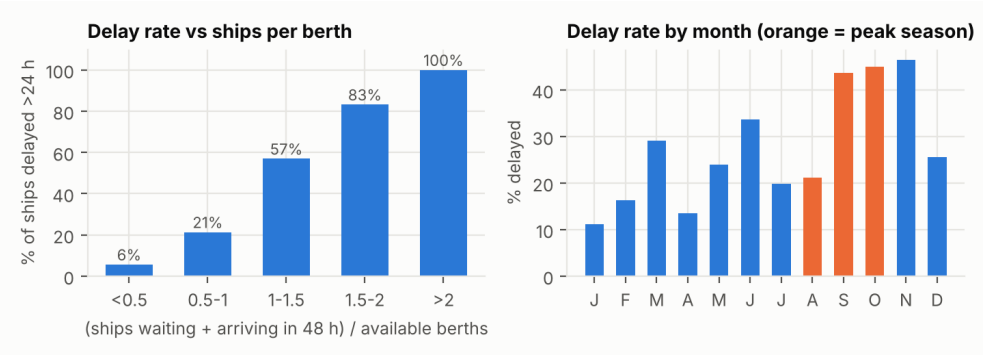


Figure 1. Sanity check of the simulated ground truth: delay risk climbs steeply once more than one ship competes for each berth, and is highest in peak season and cyclone season.

3. Features (all available 48 h before arrival)

Group	Features	Why it matters
Traffic & queue	ships waiting now, ships due in 48 h (from ETAs), **ships per available berth**, berth occupancy	The core signal is the *interaction* of traffic and capacity
Capacity	berths available, maintenance flag	Same traffic, fewer berths → queue
Weather	forecast max wind, **forecast hours of closure**, closure-days × ships per berth	Duration matters more than peak; storms hurt more at a busy port
Labour / vessel	strike announced; ship size, needs deep-water berth	Stoppages; competition for few deep berths

Timing	peak season, night arrival, **night arrival at a daylight-only port**, weekend	Temporal patterns and port rules
History / port	average wait at this port in the last 7 days; port identity	Momentum; port baseline

4. Approach

- **Three approaches compared:** (a) a planner's rule: alert if more than 1 ship per berth, a strike is announced, or a storm is forecast at a busy port; (b) logistic regression; (c) gradient boosting.
- **Time-based evaluation, never a random split** (a random split would let the model learn from the future). Development: Nov 2024–Apr 2026. **Test: May–Oct 2026**, deliberately the hardest season (monsoon + peak), used once at the end. A ship enters training only once its outcome is known.
- **Model choice by walk-forward validation:** 3 folds (Aug–Oct 2025, Nov–Jan, Feb–Apr), each trained only on the past. Logistic regression won on average (ROC-AUC 0.944 vs 0.913 for boosting and 0.914 for the rule; Brier 0.085 vs 0.099). The key interactions are engineered as features, so a linear model can use them, and with ~1,500 rows boosting overfits slightly. The simpler model is also easier to explain and maintain.
- **Alert threshold = 25%, from decision theory rather than tuning.** If a missed delay costs 3× a false alarm, alerting pays off when P×3 > (1−P)×1, i.e. P > 1/(1+3). The validation cost curve is flat around it (best 0.19: 20.4 vs 22.4 per 100 ships), so the principled value was kept to avoid overfitting.
- **Confidence = share of 25 model versions that reach the same alert decision.** Each version is trained on a different bootstrap resample and fills any missing input with a different plausible value from that port's history. One number captures three doubts: the model is unsure, the probability sits near the alert line, or key data is missing. Probability says *how likely* the delay is; confidence says *how stable* that answer is.
- **Explanations by "what if this factor were normal?":** each factor group is reset to a typical day at that port and the model re-predicts. This works for any model, and every number can be said out loud and checked. An **hours estimate** comes from past ships at the same port with similar predicted risk and how long they actually waited.

5. Results

Approach (test: May–Oct 2026)	ROC-AUC	PR-AUC	Recall	Precision	F1	Brier	Cost / 100 ships*
Never alert	0.50	–	0%	–	0.00	–	101
Always alert	0.50	–	100%	34%	0.50	–	66
Planner's rule (baseline)	0.893	0.791	77%	76%	0.76	–	31.7
Logistic regression (chosen)	**0.924**	**0.861**	**81%**	**79%**	**0.80**	**0.105**	**26.2**
Gradient boosting	0.916	0.846	81%	78%	0.79	0.119	27.0

*Cost = 3 × missed delays + 1 × false alarms per 100 ships. Test delay rate 34%. Out of 177 real delays, 144 were caught; 39 false alarms among 349 on-time ships.

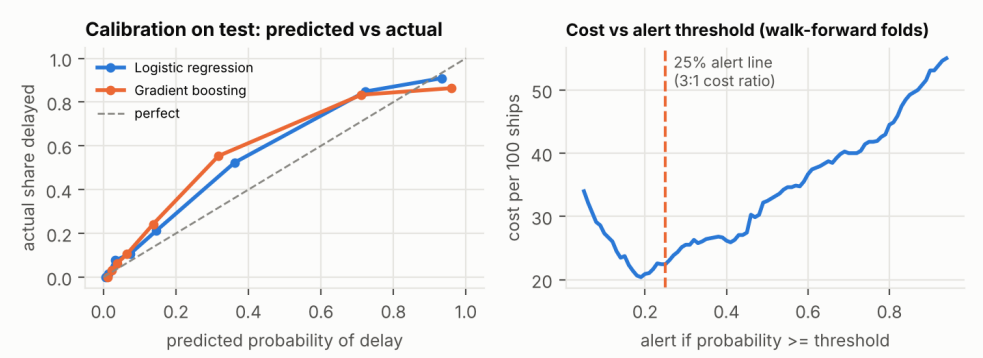


Figure 2. Left: calibration on the test period. Probabilities are reliable at the extremes, but mid-range risk is **underestimated** in monsoon/peak season (seasonal drift, see Section 7). Right: cost per 100 ships vs alert threshold across walk-forward folds; the 25% line sits in the flat region.

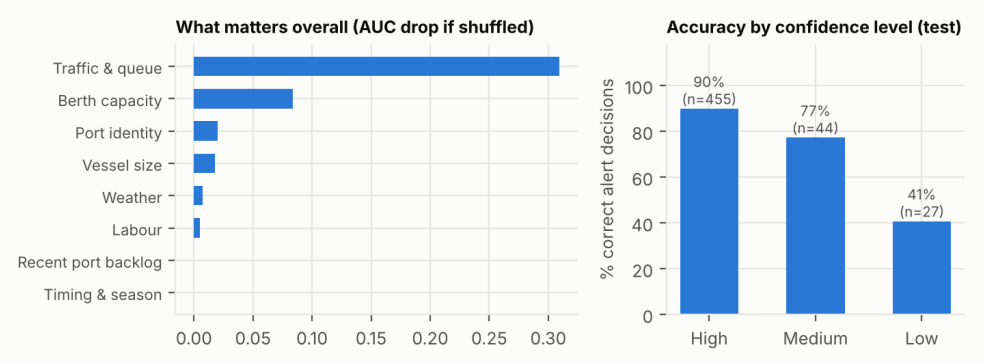


Figure 3. Left: global importance (drop in test ROC-AUC when a factor is shuffled). Right: the confidence score separates reliable from unreliable predictions.

- **The confidence score is meaningful:** High-confidence calls (87% of cases) are correct 90% of the time, Medium 77%, Low 41%. Low is close to a coin flip, so the system correctly flags it.
- **Traffic per berth dominates** (AUC drop 0.31), then berth capacity (0.08). Weather and strikes rank low *on average* because they affect few ships, but they are decisive when they occur: when a storm hit during the wait, 86% of ships were delayed. Global importance and per-ship explanations answer different questions.
- **Per port** ROC-AUC ranges from 0.88 (Northbay) to 0.95 (Southreach). Northbay has the lowest recall (64%) because delays there are rare (12%) and come from short traffic waves.
- **Error analysis** of the 33 missed delays: 16 borderline, 13 queue built up faster than expected, 3 storm not in forecast / forecast missing, 1 incoming-traffic data missing. 48% were borderline (waited 24–36 h); median wait of missed delays 37 h.

6. From prediction to operational decision

Risk is shown in three bands with a matching action: **Low** (<25%): proceed; **Elevated** (25–60%): warn the customer and re-check at 24 h; **High** (≥60%): move the delivery date and re-book trucks/rail. The advice depends on the *main driver*. Traffic → slow-steam to arrive just in time (same berth time, less fuel and anchorage cost) or compare an alternative port; weather → expect a 1–2 day slip, don't re-route; strike → line up a backup port; maintenance → request a berth window after it reopens. A real test case, predicted 48 h ahead:

PC01616 · Eastgate · arrives 31 May 2026 09:00
Delay risk **75% (High)** · Confidence **High** (25/25 model versions agree)
• **Traffic & queue**: 3 of 3 berths busy now; 3 waiting + 1 due in 48 h for 3 berths = 1.3 ships per berth (normal here: 0.7) → +45 pts (risk if normal: 29%)
• **Weather**: storm forecast - about 23 h of port closure expected (winds up to 78 km/h; port stops above 45) → +23 pts (risk if normal: 51%)
• Similar-risk past ships here waited a median of 50 h (typical 32–73 h)
• **Action:** move delivery date and re-book transport; slow-steam to arrive near the likely berth time; treat the storm part as a short slip, not a reason to re-route.
What happened: the ship waited 83 h.

7. Assumptions and limitations

- **Synthetic data is the main limitation.** The model can only learn what the simulator contains; real ports add customs holds, yard congestion, crane breakdowns and priority berthing for liner contracts. The results show that the *method* works end to end, not how accurate it would be at a real port.
- **Seasonal drift:** on the monsoon/peak test period the model underestimates mid-range risk (Figure 2). In production: recalibrate monthly and track calibration by season and port.
- **One-factor-at-a-time explanations** can understate a factor when another already pushes risk near 100% (saturation). SHAP values would split credit more fairly; they were left out to keep dependencies minimal.
- **Assumed numbers:** the 3:1 cost ratio, the 24 h line and the 48 h horizon should come from the business. All are settings, and the alert threshold follows the cost ratio automatically.
- **Ports are simulated independently**; in reality ships divert between neighbouring ports. Ship ETAs and forecasts are noisy but unbiased here; real feeds can be systematically late.

8. Next steps with real data

Using AIS data (e.g. MarineCadastre): detect port calls by finding ships that stop inside an anchorage polygon (speed < 1 knot) and later enter a berth polygon. Waiting time = berth entry − anchorage entry, and queue length and arrivals-in-48h come from the same feed. Then add weather forecasts and port notices, and keep the same pipeline: features → walk-forward validation → calibrated probability → confidence → explanation → action. A per-port model or recalibration and a feedback loop (planners marking alerts as useful or not) would follow.